Constraint-Based Modelling for the study of Polymicrobial Diseases

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Abstract

Polymicrobial diseases are caused by multiple pathogens concurrently infecting a specific nice in a host. While Genome-Scale Metabolic Models (GEMs) have been used to study the metabolic interactions between a single pathogen and an infected host cell, models that integrate host GEMs with those of multiple coinfecting pathogens have not been developed. This study aims to develop an integrated model to capture the host-pathogen metabolic interactions in the polymicrobial disease caused by *Vibrio cholerae* and Enterotoxic *Escherichia coli* (ETEC) in a murine model of small intestine epithelial cells, or enterocytes.

Introduction

Polymicrobial Diseases are the diseases caused by multiple pathogens infecting a specific cell or tissue type within a host at the same time. Multiple polymicrobial diseases, both acute and chronic, and occurring across multiple niches, have been characterised in humans [1].

In certain cases, the infection by one pathogen creates a niche that favours infection and colonisation by another pathogen; such interactions are referred to as **synergistic polymicrobial infections**. For instance, the metabolic interactions between multiple bacteria cause periodontal diseases are known to play a synergistic effect [1].

In other cases, the presence of one pathogen may also inhibit colonisation by another pathogen; such interactions are referred to as **microbial interference**. For example, several studies indicate a negative correlation between the microbial loads of *S. pneumoniae* and *S. aureus* due to bacterial interference [1].

Such interactions between different microbes in a host niche can be both **metabolic** (for example, certain processed macromolecules secreted by one pathogen can be taken up by another) as well as **non-metabolic** (for example, those involving interactions mediated by bacterial protein secretion systems) in nature [1].

The present study aims to characterise the metabolic interactions between *Vibrio cholerae* and Enterotoxic *Escherichia coli* (ETEC) co-infecting the enterocyte niche in a **murine host model** using Genome-Scale Metabolic Models (GEMs).

Motivation

Previous studies have tried to capture host-pathogen metabolic interactions in integrated GEMs comprising of the metabolic networks of a host and a single infecting pathogen. For instance, such interactions have been extensively studied in the case of the infection of *M. tuberculosis* in alveolar macrophages [2].

However, there are no reports of integrated GEMs that take into account the **metabolic networks of more than one pathogen**, which would be ideal for characterising polymicrobial infections.

Therefore, this study aims to integrate the metabolic models of V. cholerae, ETEC and murine enterocytes to comprehensively study the effect of polymicrobial diseases in this specific niche.

Methods

Manually curated GEMs of the pathogens V. cholerae and ETEC, as well as an integrated GEM comprising of both the pathogens were obtained from a previous study [3]. This study explored the interactions between the two pathogens by comparing the performance of their individual GEMs and the integrated metabolic model in different media in terms of their growth capabilities and gene essentiality predictions.

Since GEMs representing human enterocytes were not readily available, a GEM representing **murine enterocytes** was obtained from a previous study [4]. Since this model relied on data from human enterocytes to set upper and lower bounds for exchange reaction fluxes, the same constraints were used for the purposes of this study.

The cobrapy package in Python was used to import and read the relevant models, and perform the required analyses.

Data Pre-Processing

Some of the initial challenges pertaining to data pre-processing were handled using the following methods.

Since the GEMs relevant for this study were obtained from different sources, multiple steps of data pre-processing had to be performed to ensure that subsequent analyses do not raise **SBML or cobrapy errors** and that coherent results are obtained on further analyses.

The GEMs obtained from different sources were subjected to **model validation**, using the relevant function provided by cobrapy. Accordingly, respective GEMs were modified to eliminate the errors raised during the model validation step.

For instance, the metabolite chitin in the V. cholerae GEM was modified to reflect its monomer, since a polymeric formula $H_2O(C_8H_{13}NO_5)_n$, consisting of 'n' monomer units, was inconsistent with SBML rules that require chemical formulae to only consist of strings comprising either atomic names or user defined compounds.

Furthermore, the **metabolite labels** across all models were made consistent with respect to the ones used in the murine enterocyte model, since this model consists of more intracellular compartments as compared to the other pathogen models.

The models were considered to be suitable for further analyses, including the integration of their stoichiometric matrices, as described elsewhere [2, 3].

Results

Alongside data pre-processing, some of the results pertaining to gene essentiality analyses and growth capabilities of the individual pathogen models and the co-culture model, as described in [3], were replicated to understand the models better, as well as to develop familiarity with the cobrapy package and different data visualisation techniques.

Future Work

After replicating the **gene essentiality** and **growth capabilities analyses** for the individual pathogen models and the co-culture model (from [3]), we aim to perform similar analyses for the murine enterocyte model and the integrated host-pathogen model, and compare the predictions across different models.

This will help in understanding the differences in the metabolic capabilities of host and pathogen cells before and after infections. Furthermore, identifying essential genes for one or more pathogens in the integrated host-pathogen model can provide new targets for the treatment of polymicrobial diseases.

Furthermore, we aim to perform **dynamic FBA**, which is typically used to study microbial cultures, on the integrated host-pathogen model to understand the temporal dynamics of polymicrobial infections.

Challenges

A challenge in host-pathogen modelling for polymicrobial diseases is the lack of *in vivo* or *in vitro* transcriptome data for all the different cell types involved. Therefore, accurately setting flux constraints for individual reactions remains difficult during such modelling activities.

In this case, the transcriptome data obtained from a V. cholerae and ETEC co-culture is available [3]. This can be considered as a starting point to set up flux constraints for relevant pathogen-associated reactions in the integrated host-pathogen model, and subsequently subject the model to different analyses, as described above.

Conclusion

We have developed an integrated host-pathogen model for the co-infection of V. cholerae and ETEC in murine enterocyte cells.

We further aim to perform several analyses of interest and thereby utilise the integrated model to explore the metabolic interactions between the host and different pathogens that underlie the pathogenesis of a specific polymicrobial disease.

References

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